Amlak Finance PJSC

Procure-to-Pay (P2P) Engine & Central Approval HUB

System Architecture, End-to-End Workflow & Technical Implementation Blueprint

Target Platform: Amlak ORBIT Portal	**Version:** 1.0 (Enterprise Specification)	**Date:** October 2026
Primary Stakeholders: Procurement, Finance, Management	**Integration:** Oracle EBS & Central HUB	**Security:** Microsoft Entra ID (SSO)

1. Executive Summary & Strategic Objectives

Amlak Finance PJSC currently relies on disparate SharePoint evaluation forms and standalone Oracle E-Business Suite (EBS) modules for its purchasing lifecycles. This proposal outlines the strategy to integrate a unified, modern **Procure-to-Pay (P2P) Workflow Engine** and an enterprise-grade **Central Approval HUB** directly into the Amlak ORBIT ecosystem.

Core Design Principles:

- **Preserve Existing ERP Investment:** The solution does not replace Oracle EBS; it bridges the gap by modernizing pre-PO sourcing and creating a lightweight EBS integration boundary.
- **Centralized Multi-Tier Governance:** Managers and executive authorities review all organizational approvals (RFP, Sourcing, Budget, Payment) in a single unified inbox.
- **Fraud Prevention via 3-Way Matching:** Automatic programmatic verification comparing Purchase Order (PO) terms, Goods Receipt Notes (GRN), and Vendor Invoices before funds disbursement.

2. Amlak's 12-Step Procurement Lifecycle

The end-to-end procurement cycle is divided into three distinct operational phases across 12 sequential milestones:

Step	Phase	Process Milestone	Operational Activity & Key Data Captured
1	Phase A	**RFP Creation**	Department defines requirement, specifications, budget estimate, and target timeline.
2	Phase A	**RFP Approval**	Department Head verifies operational justification and authorizes public/private tender.
3	Phase A	**Vendor Shortlisting**	Submissions received; preliminary filtering filters out non-compliant bidders.
4	Phase A	**Vendor Scoring**	Systematic weighted evaluation: Technical (e.g. 70%) and Commercial/Cost (e.g. 30%).
5	Phase B	**Procurement Request**	Formal requisition submitted attaching top-ranked vendor proposal & business case.
6	Phase B	**Procurement Approval**	Procurement Committee / Delegated Authority formally authorizes vendor selection.
7	Phase B	**Approved Procurement**	Decision finalized, audit timestamped, and staged for binding financial commitment.
8	Phase B	**Purchase Order in EBS**	Official legal PO generated in Oracle EBS format with unique PO Reference Number.
9	Phase C	**Finance Budget Check**	Finance Controller verifies GL cost center allocations and budget availability.

10	Phase C	**Fulfilment & Invoice**	Vendor delivers goods/services; Store confirms delivery (GRN) and vendor submits bill.
11	Phase C	**Payment Request**	Accounts Payable submits digital payment packet upon 3-Way Match validation.
12	Phase C	**Payment Approval**	Authorized Finance Signatory approves release of funds (Wire/EFT/Cheque).

3. The ORBIT Central Approval HUB Architecture

A critical bottleneck in multi-departmental enterprises is 'Approval Fragmentation', where managers log into multiple emails and isolated software suites to authorize tasks. The **ORBIT Central Approval HUB** provides a single, unified executive inbox.

ORBIT CENTRAL APPROVAL HUB (Unified Workflow Inbox)

Checkpoint 1: RFP Sourcing Approval | **Checkpoint 2:** Vendor Recommendation | **Checkpoint 3:** Budget Clearance | **Checkpoint 4:** Final Payment Signoff

Decoupled Data Architecture: The HUB stores workflow states, assigned approvers, timestamps, and audit notes. The underlying business payload (RFP specs, PO contracts, tax invoices) remains governed by the respective functional modules.

4. Technical Implementation & Integration Blueprint

To ensure seamless compatibility with the existing Amlak Finance infrastructure, the procurement engine will extend the existing NestJS and React foundation:

Layer	Component	Technical Architecture & Functionality
Frontend UI	**React 19 + Vite + Tailwind CSS**	• Responsive multi-step Procurement Stepper tracking Phase A, B, and C. • Interactive Vendor Scoring Matrix with real-time ranking recalculation. • Central Approval HUB inbox view with 1-click Approve/Reject modals.
Backend API	**NestJS Modular Microservice**	• ProcurementModule: Endpoints for RFP, Vendor proposals, and scoring. • ApprovalsModule: Unified queue management, status transitions, notifications. • InvoicesModule: Automated 3-way matching engine.
Data Layer	**Prisma ORM (Postgres / SQLite)**	• Type-safe schema models: Rfp, VendorProposal, PurchaseOrder, ApprovalTask, Invoice. • Zero-config local portability (SQLite) with instant enterprise Postgres production switch.
ERP Bridge	**Oracle EBS Integration Boundary**	• Generates structured EBS-compliant PO payloads. • Prototype simulates PO reference numbers (e.g. AMLAK-PO-2026-XXXX) with bidirectional status mapping for future web services/REST integration.
Identity / IAM	**Microsoft Entra ID (SSO) + 2FA**	• Unified login using corporate Microsoft accounts (@amlakfinance.com). • Strict Role-Based Access Control (Admin, Requester, Technical Evaluator, Finance Approver).

5. Automated 3-Way Matching Mechanism (Anti-Fraud Control)

Prior to disbursing vendor payments, the system executes an automated 3-way reconciliation audit:

Record Source	Key Parameters Verified	Reconciliation Rule
1. Purchase Order (PO)	Authorized unit cost, currency, quantity, payment terms	Establishes approved commercial limits.
2. Goods Receipt (GRN)	Actual physical quantity delivered, condition inspection pass	Prevents paying for unreceived/damaged items.
3. Vendor Invoice	Billed item quantity, price per unit, sales taxes, bank IBAN	Invoice line items must strictly match PO & GRN.

6. Implementation Roadmap & Phased Delivery Milestones

Milestone Phase	Target Timeline	Deliverables & Core Capabilities
Phase 1: Sourcing & Scoring	Week 1 - 2	RFP submission portal, vendor quotation ingestion, and multi-parameter scoring matrix (Technical vs Commercial calculation).
Phase 2: Central Approval HUB	Week 3 - 4	Unified multi-stage approval queue, audit history logging, delegation rules, and email/toast action notifications.
Phase 3: EBS PO & 3-Way Match	Week 5 - 6	EBS PO simulator, delivery receipt logger (GRN), invoice parser, and automated 3-way match validation engine.
Phase 4: User Testing & Hardening	Week 7	End-to-end integration testing with Amlak Finance stakeholders, security audit, and production deployment readiness.

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